

Personalized azithromycin treatment rules for children with watery diarrhea using machine learning

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Personalized Azithromycin Treatment Rules for Children with Watery Diarrhea Using Machine Learning

The authors propose a rule to identify children who would benefit most from antibiotic treatment for acute diarrhea. This study is a secondary data analysis of the ABCD trial.

While the benefits in risk reduction for the recommended groups are statistically significant, they are small. I have several methodological concerns and suggestions for improvement:

Practical Applicability: While the authors demonstrate that risks of outcomes differ between groups, this alone does not prove the rule's practical utility. The lack of a test set, likely due to the limited number of cases, is a significant limitation of this approach.

Overlap in Patient Selection: The authors report benefits of personalization for specific outcomes such as 3-day diarrhea resolution and rehospitalization/death. However, it would be valuable to examine whether the patients selected for treatment overlap across these outcomes. If the rule does not consistently select the same patients who benefit, this would indicate low robustness. A Venn diagram visualizing this overlap would be helpful.

Performance Estimation: The use of 10-fold cross-validation yields performance metrics as averages across folds, which is not ideal for robust performance estimation. The authors should elaborate on the statistical basis for computing confidence intervals in this setting.

Generalizability: The authors have not demonstrated that their results are generalizable. It remains unclear how the model would perform on data from countries not included in the training set, raising concerns about its broader applicability.

Inconsistencies in Reporting: In line 142, the percentage of children with recommendations (27.8%) differs from the value reported in Table 1. Similar discrepancies are noted for other outcomes and should be clarified.

Explainability of the Rule: The proposed rule lacks explainability, which limits its interpretability and potential adoption in clinical settings.

Comparison with Existing Guidelines: In line 84, the authors mention that WHO recommends antibiotics for cases of bloody stool. It would be helpful to compare the performance of their comprehensive rule with the WHO guidelines to assess added value.

Conclusion:

Overall, the paper's significance is limited due to the small effect size and reliance on cross-validation with averaged results across folds to achieve statistical significance. This suggests weak evidence for the robustness and generalizability of the proposed methods. Addressing the concerns outlined above would strengthen the study's impact and practical relevance.

(Remarks on code availability)

Code looks very nice, although I have not tried it as

Reviewer #2

(Remarks to the Author)

Relevant study. Important field.

ABSTRACT reads well. If possible try to clarify what is machine-learning-based procedure already in abstract. Otherwise good abstract with absolute numbers, percentages, 95%CI. Relevant outcomes, Authors present study limitations already in the abstract.

1. The one thing I suggest the authors change is that I wonder if the $n=6,692$ is a bit misleading? in the main text it seems a much smaller number of children were included in the analysis for this paper.

2. Please also present "number needed to treat":

- to avoid diarrhoea on day 3

- to avoid 1 hospitalisation or death.

3. since this was a treatment study, please also add one sentence on side effects (they are especially important) since the absolute benefit of the study may be limited?

Introduction well written. motivates the study. Diarrhea is a global threat.

interesting that you use both symptoms and pathogen to decide who will receive antibiotics (and I agree some kind of rule is needed since the overall trial was negative)

Results:

4. once again, clarify the actual number of children in the study? 6692??

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6, please explain the concordance correlation coefficient and its meaning.

7, how does the machine learning actually work in this study?

Discussion

8. I think the reasoning of the discussion is ok including the argument that azithromycin may be anti-inflammatory.

I think the authors present their limitations in an appropriate way

Methods:

9. the overall methods are well described.

(Remarks on code availability)

-

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

I have no further comments.

(Remarks on code availability)

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REVIEWER COMMENTS
Manuscript ID: NCOMMS-24-69871

Reviewer #1 (Remarks to the Author):

Personalized Azithromycin Treatment Rules for Children with Watery Diarrhea Using Machine Learning

The authors propose a rule to identify children who would benefit most from antibiotic treatment for acute diarrhea. This study is a secondary data analysis of the ABCD trial.

While the benefits in risk reduction for the recommended groups are statistically significant, they are small. I have several methodological concerns and suggestions for improvement:

Practical Applicability: While the authors demonstrate that risks of outcomes differ between groups, this alone does not prove the rule's practical utility. The lack of a test set, likely due to the limited number of cases, is a significant limitation of this approach.

We agree that demonstration of applicability using an external dataset, or test set, better proves the rule's practical utility. Therefore, we have added a new section "v. Application to External Dataset" in the results of the revised text for the application of the rules learned in the ABCD Trial data to an external test set, data from the Global Enteric Multicenter Study (GEMS). The new section in lines 289-303 states:

"To demonstrate the generalizability and practical applicability of developing personalized treatment rules, we applied ABCD-learned rules for antibiotic treatment of day 3 diarrhea to the Global Enteric Multicenter Study (GEMS), which was a 7 site, case-control study for acute moderate-to-severe-diarrhea conducted between 2008 - 2011. Under the ABCD-learned comprehensive rule, the treatment effect across 5 seeds among children recommended for treatment in GEMS was -0.058 (95% CI: -0.189, 0.072) and among children not recommended for treatment was -0.008 (95% CI: -0.083, 0.067) (S. Table 7). The pathogen rule and host rule were also able to reasonably discern which children would benefit more from antibiotic treatment in GEMS. Similar to ABCD results, the child-level expected benefit under the pathogen rule applied to GEMS was correlated with that of comprehensive rule (CCC:0.69; 95% CI: 0.67, 0.71). Despite GEMS being non-randomized, including different study sites, and occurring 6 years prior, the rules learned in ABCD data were still able to identify heterogeneity in treatment effects against diarrhea duration in this external dataset."

S. Table 7. Proportion of watery diarrhea episodes treated with azithromycin and average benefits of personalized treatment rules for day 3 diarrhea averaged across 5 seeds in GEMS under ABCD-learned rules

	Proportion Treated (95% CI)	Average Benefit among Recommended for Treatment (95% CI)	Average Benefit among Not Recommended for Treatment (95% CI)

Comprehensive Rule	0.242 (0.225, 0.258)	-0.058 (-0.189, 0.072)	-0.008 (-0.083, 0.067)
Pathogen Rule	0.291 (0.273, 0.308)	-0.054 (-0.147, 0.039)	-0.016 (-0.099, 0.067)
Host Rule	0.261 (0.244, 0.278)	-0.074 (-0.229, 0.081)	-0.010 (-0.077, 0.058)

Overlap in Patient Selection: The authors report benefits of personalization for specific outcomes such as 3-day diarrhea resolution and rehospitalization/death. However, it would be valuable to examine whether the patients selected for treatment overlap across these outcomes. If the rule does not consistently select the same patients who benefit, this would indicate low robustness. A Venn diagram visualizing this overlap would be helpful.

We agree these are important data to show and have added a 2x2 table accordingly in S Table 8. However, we do not expect the treatment rules would necessarily recommend treatment to the same children since the determinants of proximal outcomes like diarrhea duration (e.g., pathogen causing the illness) may be different than the determinants of longer-term severe outcomes like death (e.g., child malnutrition status). We discuss the reasons for this in paragraph 335-354 of the revised text. In lines 351-353 of the discussion section, we added a 2x2 table for the overlap in patient selection for treatment for the two outcomes under the comprehensive rule with the following explanation to clarify this difference:

“Given that mechanisms by which antibiotics affect diarrhea duration and re-hospitalization or death could be distinct, we would not expect a single treatment rule to identify the same children across different outcomes (S. Table 8).”

S. Table 8. Distribution of children recommended for treatment under the comprehensive rule using the seed with median CATE for day 3 diarrhea and day 90 re-hospitalization or death.

	Decision to treat for day 90 re-hospitalization or death		
Decision to treat for day 3 diarrhea	No	Yes	Total
No	3323	1679	5002
Yes	930	760	1690
Total	4253	2439	6692

Performance Estimation: The use of 10-fold cross-validation yields performance metrics as averages across folds, which is not ideal for robust performance estimation. The authors should elaborate on the statistical basis for computing confidence intervals in this setting.

Thanks for the comment. We agree that cross-validated performance measures are not always ideal in that they report performance averaged across multiple models.

Nevertheless, we believe that the approach is supported by recent studies in the statistical literature. A discussion for the basis of cross-validation-based inference can be found in Hubbard et al (doi: 10.1515/ijb-2015-0013). There the authors propose influence function-based standard error estimates that are used to construct confidence intervals – the same approach we used in this work. Moreover, other recent work done by Benkeser et al (doi: 10.1080/01621459.2019.1668794) suggests that inference regarding cross-validation-based performance metrics for a given prediction algorithm is approximately valid for inference regarding true hold-out performance on an algorithm applied to the full data set in large samples. Finally, recent work related specifically to treatment rule estimation has found that estimation error for the average outcome under the optimal treatment rule is identical in first-order to that of estimating the average outcome under a fixed treatment rule (e.g., one learned via cross-validation) (see van der Laan and Luedtke doi: 10.1515/jci-2013-0022). This suggests that cross-validated evaluation of the performance of a treatment rules likely provides a reasonable approximation to the estimated performance of the true optimal treatment rule given a particular set of variables.

Overall, we believe that the internal validity of our results is reasonably well established via cross-validation and the addition of the independent, validation data (see above) supports the external validity of our approach as well.

We have added the following summary to the revised text to provide additional justification for our approach in lines 498-500 with three citations included for reference:

“Cross-validated evaluation of the performance of treatment rules likely provides reasonable inference and approximation to the estimated performance of the true optimal treatment rule given a particular set of values (Hubbard, Benkeser, van der Laan and Leudtke).”

Generalizability: The authors have not demonstrated that their results are generalizable. It remains unclear how the model would perform on data from countries not included in the training set, raising concerns about its broader applicability.

The reviewer makes an important point about demonstrating generalizability. To demonstrate generalizability, we applied the ABCD treatment rules to GEMS, which was conducted 6 years earlier, is an observational study, and included study sites in countries that were not in ABCD, including the Gambia and Mozambique. The addition of this application has been described above for the first reviewer comment and has been added to the manuscript in lines 289-303 of the revised text.

Inconsistencies in Reporting: In line 142, the percentage of children with recommendations (27.8%) differs from the value reported in Table 1. Similar discrepancies are noted for other outcomes and should be clarified.

The percentage of children recommended for treatment in the results text differs from that in Table 1 because we used the average across seeds for the treatment effects while we used the treatment distribution from the seed with the median predicted expected benefit for the descriptive analyses. In the methods section of the revised text, we added the following language in lines 507-510 to clarify the use of the median seed for descriptive results:

“Because results were repeated across five seeds, we presented the distribution of characteristics based on treatment recommendation from the seed with the median expected child level benefit for the descriptive analyses.” We additionally added a foot note to Table 1 to further clarify the discrepancy.

Explainability of the Rule: The proposed rule lacks explainability, which limits its interpretability and potential adoption in clinical settings.

While we agree with the reviewer that the rules do not provide interpretability, our comparison of rules derived on different subsets of data still provides value in identifying which data (e.g. symptoms, demographics, anthropometry, diagnostics, etc.) are the strongest determinants of treatment benefit. We have clarified our intention and value of this aim by revising and more explicitly adding language to the last paragraph of the introduction of the revised text in lines 125-129 to read:

“Through this process, we aim to interrogate whether certain types of data describing a child or their illness would be important for determining antibiotic treatment benefit. If data such as clinical or sociodemographic characteristics can sufficiently capture heterogeneity in the azithromycin treatment response, appropriate targeting of antibiotic treatment may be achievable even when point-of-care diagnostics are unavailable.”

We additionally added the following sentence to the discussion section of the revised text in lines 319-321:

“Furthermore, the rule could be adopted in clinical settings using a mobile based application, for example, without requiring the reasons for treatment recommendations to be completely interpretable.”

Comparison with Existing Guidelines: In line 84, the authors mention that WHO recommends antibiotics for cases of bloody stool. It would be helpful to compare the performance of their comprehensive rule with the WHO guidelines to assess added value.

We agree that this would be a valuable comparison, but our research question focuses on the value of antibiotics for watery diarrhea. Furthermore, we are unable to develop a rule based on the WHO guidelines because the ABCD Trial excluded children with bloody stool due to lack of equipoise in that subgroup. In lines 392-396 of the revised text, we have added the reviewer’s suggestion for future direction:

“Because the ABCD Trial excluded children with bloody stool, we were unable to compare the performance of the rules with the WHO guidelines to evaluate their

potential added value to the current recommendations. One important avenue for future research could involve comparing the performance of these treatment rules to the WHO guidelines in datasets where relevant data are available.”

Conclusion:

Overall, the paper’s significance is limited due to the small effect size and reliance on cross-validation with averaged results across folds to achieve statistical significance. This suggests weak evidence for the robustness and generalizability of the proposed methods. Addressing the concerns outlined above would strengthen the study’s impact and practical relevance.

While the effect sizes are modest relative to the precision of estimates, the number needed to treat is quite low (under the comprehensive rule, treat approximately 10 children recommended for treatment to expect 1 fewer case of diarrhea lasting until day 3 and treat approximately 42 children recommended for treatment to expect 1 fewer case re-hospitalization or death by day 90), suggesting we are capturing public health relevant effects. We agree that the modifications based on the concerns listed above have strengthened the study’s impact and practical relevance, particularly the validation in a completely external data source.

Reviewer #1 (Remarks on code availability):

Code looks very nice, although I have not tried it as

Reviewer #2 (Remarks to the Author):

Relevant study. Important field.

ABSTRACT reads well. If possible try to clarify what is machine-learning-based procedure already in abstract. Otherwise good abstract with absolute numbers, percentages, 95% CIs. Relevant outcomes, Authors present study limitations already in the abstract.

We clarified the role of the machine learning-based procedure by adding the following language to the abstract in lines 53-55 of the revised text:

“This procedure estimates the child-level expected benefit for a given set of covariates by combining predictions from a library of statistical models.”

1. The one thing I suggest the authors change is that I wonder if the n=6,692 is a bit misleading? in the main text it seems a much smaller number of children were included in the analysis for this paper.

We recognize that only the supplementary methods explained how children who were missing outcomes were retained in the analysis so that all 6,692 children were included. In lines 140-142 of the revised main text, we added the following language to clarify the process in which 6,692 children were retained in the analysis despite missing outcomes:

“...via a nuisance regression that modeled the probability of having a non-missing outcome given azithromycin treatment assignment and all measured covariates. (S. Methods). Therefore, we retained 6,692 children in the analysis.”

2. Please also present “number needed to treat”:

We agree with the reviewer that presenting the number needed to treat would increase the interpretability of the treatment effects. We added the following language to include number needed to treat for the treatment effects among those recommended for treatment and not recommended for treatment under the comprehensive rule:

- to avoid diarrhoea on day 3

In lines 151-153: For the NNT under the comprehensive rule among those recommended for treatment: “...translating to a number needed to treat (NNT) of approximately 10 children to expect 1 fewer case of diarrhea lasting until day 3 post-treatment.”

In line 154: NNT under the comprehensive rule among those not recommended for treatment: “(NNT: 27)”

- to avoid 1 hospitalisation or death.

In lines 196-198: For the NNT under the comprehensive rule among those recommended for treatment: “...translating to a number needed to treat of approximately 42 children to expect 1 fewer instance of re-hospitalization or death in the 90 days following treatment”

In line 199: NNT under the comprehensive rule among those not recommended for treatment: “(NNT: 91)”

3. since this was a treatment study, please also add one sentence on side effects (they are especially important) since the absolute benefit of the study may be limited?

We agree with the reviewer that side effects may influence determination of the most appropriate rule. Therefore, we have added language on side effects in lines 328-335:

“Deciding on the most appropriate personalized treatment rule in a given setting should weigh performance of the algorithm against feasibility with respect to data availability

and limits on antibiotic use such as antibiotic resistance and common antibiotic side effects (i.e., nausea, abdominal pain, or vomiting). In the ABCD trial, serious events were limited to 1 participant in the azithromycin group experiencing seizure and 45 (1.1%) children in the azithromycin group requiring hospitalization or experiencing death by day 10, which was a similar proportion observed in the placebo group. ”

Introduction well written. motivates the study. Diarrhea is a global threat. interesting that you use both symptoms and pathogen to decide who will receive antibiotics (and I agree some kind of rule is needed since the overall trial was negative)

Results:

4. once again, clarify the actual number of children in the study? 6692??

We describe our clarification of this in the reviewer’s first point of feedback.

5. fo the diarrhoea outcome, do not only give me 27.8% but also the absolute number.

We have added the absolute numbers for the proportion of children recommended for treatment across the outcomes.

For day 3 diarrhea:

- In line 146: “...27.8% (n=1860; 95% CI: 26.7%, 28.9%) of children were recommended for treatment...”
- In line 172: “While the Shigella rule recommended treatment for a smaller proportion of children (15.9% [n=1112; 95% CI: 15.0%, 16.8%])...”

For day 90 re-hospitalization or death:

- In line 192: “... on average 38.0% (n=2543; 95% CI: 36.9%, 39.2%) of children were recommended azithromycin treatment...”
- In line 219: “...but the proportion treated was small (9.4% [n=629; 95% CI: 7.7%, 9.2%])...”

For change in linear growth:

- In line 242: “...an average of 37.6% (n=2516; 95% CI: 36.5%, 38.9%) of children were recommended azithromycin treatment...”

6, please explain the concordance correlation coefficient and its meaning.

We added the following language to the methods section in lines 512-516 that better explains the concordance correlation coefficient:

“... the concordance correlation coefficient which measures the level of agreement of child-level expected benefit, a continuous measure of the risk difference, obtained by two different treatment rules, the comprehensive rule and another rule. We expect that rules, that group variables into categories (Table 3), that best approximate the

comprehensive rule signal which variables may be important for determining treatment benefit.”

7, how does the machine learning actually work in this study?

In the supplemental material of the revised text, we provide a detailed accounting of the machine learning pipeline that was used:

“To further increase the flexibility of our method, we fit each regression model in the double-robust-Learner procedure using Super Learning. Super Learning employs cross-validation to combine multiple user-specified candidate regression models into an optimally weighted ensemble model. The resulting ensemble model is guaranteed to match or surpass accuracy of its individual candidate models. This approach allows us to test a wide range of models for estimating each nuisance regression and the pseudo-outcome regression, while using a data-driven approach to select between models.”

For the main text, we added the following language to the methods section in lines 455-457 to explain at baseline how the machine learning procedure works, with additional detail in the supplementary materials methods:

“The machine learning-based procedure combines predictions from a library of user-specified candidate regression models into an optimally weighted ensemble model.”

Discussion

8. I think the reasoning of the discussion is ok including the argument that azithromycin may be anti-inflammatory.

I think the authors present their limitations in an appropriate way

Methods:

9. the overall methods are well described.

Reviewer #2 (Remarks on code availability):

-